

Foundations of Cross entropy loss

[Cross entropy loss]

$$H(p, q) = -\sum_x p(x) \log q(x)$$

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Motivation In information theory, the Kraft–McMillan theorem establishes that any directly decodable coding scheme for coding a message to identify one value x_i out of a set of possibilities $\{x_1, \dots, x_n\}$ can be seen as representing an implicit probability distribution $q(x_i) = \left(\frac{1}{2}\right)^{\ell_i}$ over $\{x_1, \dots, x_n\}$, where ℓ_i is the length of the code for x_i in bits. Therefore, cross-entropy can be interpreted as the expected message-length per datum when a wrong distribution q is assumed while the data actually follows a distribution p . That is why the expectation is taken over the true probability distribution p and not q . Indeed the expected message-length under the true distribution p is

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$X \equiv (x_{11} \dots x_{1p} \dots x_{21} \dots x_{2p} \dots \dots x_{N1} \dots x_{Np}) \in \mathbb{R}^{N \times (p+1)}$, $\begin{pmatrix} x_{11} & \dots & x_{1p} & \dots & x_{21} & \dots & x_{2p} & \dots & \dots & \dots & x_{N1} & \dots & x_{Np} \end{pmatrix} \in \mathbb{R}^{N \times (p+1)}$

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Relation to linear regression The gradient of the cross-entropy loss for logistic regression is equal to the gradient of the squared-error loss for linear regression (up to a constant factor). To see this, define

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Since the logarithm is a monotonically increasing function, the maximizing value of θ is unaffected by this final step. Similarly, the maximizing value of θ is unaffected by the factor of N . So we observe that the likelihood maximization amounts to minimization of the cross-entropy.

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where we have used the calculation rules for the logarithm in the final line. Notice how the exponent contains a $-H(p, q_\theta)$ term. Taking the logarithm of both sides gives:

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$$\frac{\partial}{\partial w_j} \ln \left(\frac{1}{1 + e^{-w_j x_{ij} + k_j}} \right) = \frac{\partial}{\partial w_j} \ln \left(1 - \frac{1}{1 + e^{-w_j x_{ij} + k_j}} \right) = -\frac{x_{ij}}{1 + e^{-w_j x_{ij} + k_j}} = -x_{ij} \hat{y}_i$$

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$$H(p, q) = -\sum_m p_m \log q_m = -y \log \hat{y} - (1-y) \log (1-\hat{y})$$

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$$\frac{\partial}{\partial w_j} \ln \hat{y}_i = \frac{\partial}{\partial w_j} \ln \frac{1}{1 + e^{-w_j x_{ij} + k_j}} = x_{ij} e^{-w_j x_{ij} + k_j} \frac{1}{1 + e^{-w_j x_{ij} + k_j}} = x_{ij} (1 - \hat{y}_i)$$

$$\frac{\partial}{\partial w_j} \ln \frac{1}{1 + e^{-w_j x_{ij} + k_j}} = \frac{x_{ij} e^{-w_j x_{ij} + k_j}}{1 + e^{-w_j x_{ij} + k_j}} = x_{ij} (1 - \hat{y}_i)$$

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where the vector of weights $\mathbf{w}$ is optimized through some appropriate algorithm such as gradient descent. Similarly, the complementary probability of finding the output $y = 0$ is simply given by

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Amended cross-entropy It may be beneficial to train an ensemble of models that have diversity, such that when they are combined, their predictive accuracy is augmented. Assuming a simple ensemble of K classifiers is assembled via averaging the outputs, then the amended cross-entropy is given by

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where $\hat{y}_i \equiv g(\mathbf{w} \cdot \mathbf{x}_i) = 1 / (1 + e^{-\mathbf{w} \cdot \mathbf{x}_i})$, with $g(z)$ as the logistic function as before.

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In information theory, the cross-entropy between two probability distributions p and q , over the same underlying set of events, measures the average number of bits needed to identify an event drawn from the set when the coding scheme used for the set is optimized for an estimated probability distribution q , rather than the true distribution p .

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$$L(\theta; \mathbf{x}) = \prod_i q(\mathbf{X} = \mathbf{x}_i) = \prod_{\mathbf{x}} q(\mathbf{X} = \mathbf{x})^{\# \mathbf{x}} \approx \prod_{\mathbf{x}} q(\mathbf{X} = \mathbf{x})^N \cdot p(\mathbf{X} = \mathbf{x}) = \exp \left[\sum_{\mathbf{x}} q(\mathbf{X} = \mathbf{x}) \log \left[\frac{p(\mathbf{X} = \mathbf{x})}{q(\mathbf{X} = \mathbf{x})} \right] \right] = \exp \left[\sum_{\mathbf{x}} N \cdot p(\mathbf{X} = \mathbf{x}) \log \left[\frac{p(\mathbf{X} = \mathbf{x})}{q(\mathbf{X} = \mathbf{x})} \right] \right]$$

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NB: The notation $H(p, q)$ is also used for a different concept, the joint entropy of p and q .

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where N is the size of the test set, and $q(x)$ is the probability of event x estimated from the training set. In other words, $q(x_i)$ is the probability estimate of the model that the i -th word of the text is x_i . The sum is averaged over the N words of the test. This is a Monte Carlo estimate of the true cross-entropy, where the test set is treated as samples from $p(x)$.

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Estimation There are many situations where cross-entropy needs to be measured but the distribution of p is unknown. An example is language modeling, where a model is created based on a training set T , and then its cross-entropy is measured on a test set to assess how accurate the model is in predicting the test data. In this example, p is the true distribution of words in any corpus, and q is the distribution of words as predicted by the model. Since

the true distribution is unknown, cross-entropy cannot be directly calculated. In these cases, an estimate of cross-entropy is calculated using the following formula: